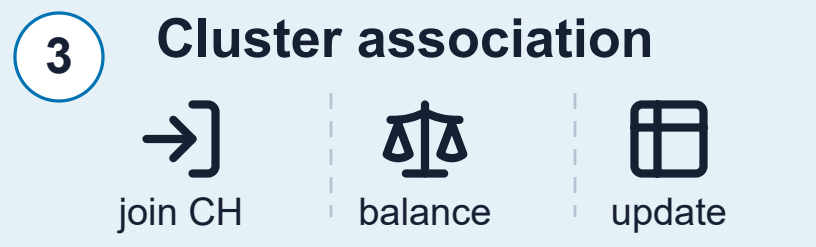
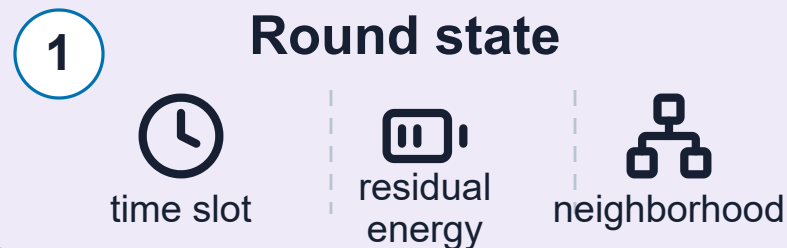
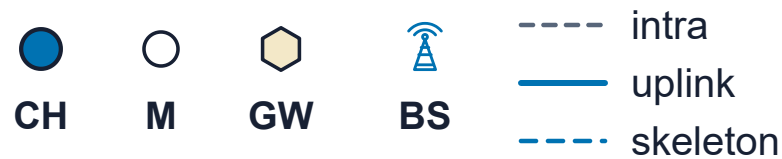
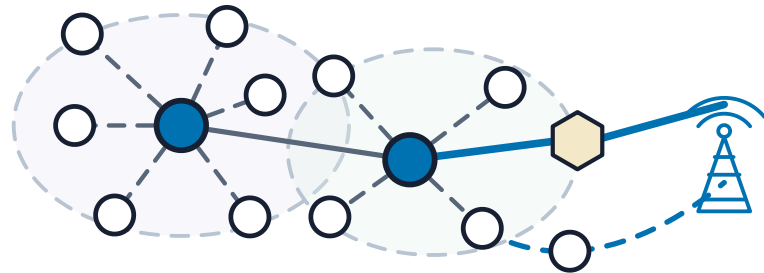




## Phase 1 Cluster Formation



## Phase 2 Intra-Cluster Communication

**Inputs:** energy · link quality · density · buffer

### 1 CAS mode selector

select one mode

**Direct**

$M \rightarrow CH$



**Chain**

$M \rightarrow M \rightarrow CH$

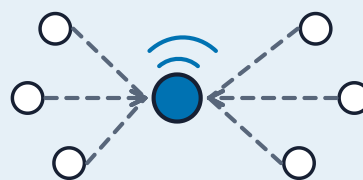


**Two-hop**

$M \rightarrow R \rightarrow CH$



### 2 Intra-cluster forwarding



### 3 CH aggregation



deduplicate



fuse



buffer



## Phase 3 Uplink to BS

*pre-transmission route selection*

